



Computer Lab - Current Lighting

This project illustrates the current lighting arrangement of the Computer Lab of Department of Electrical Engineering, University of Moratuwa

Object
Computer Lab,
Department of Electrical
Engineering,
Faculty of Engineering,
University of Moratuwa

Editor
Amasith A.T.S. - 210038K

Preface

Notes on planning:

The energy consumption quantities do not take into account light scenes and their dimming levels.

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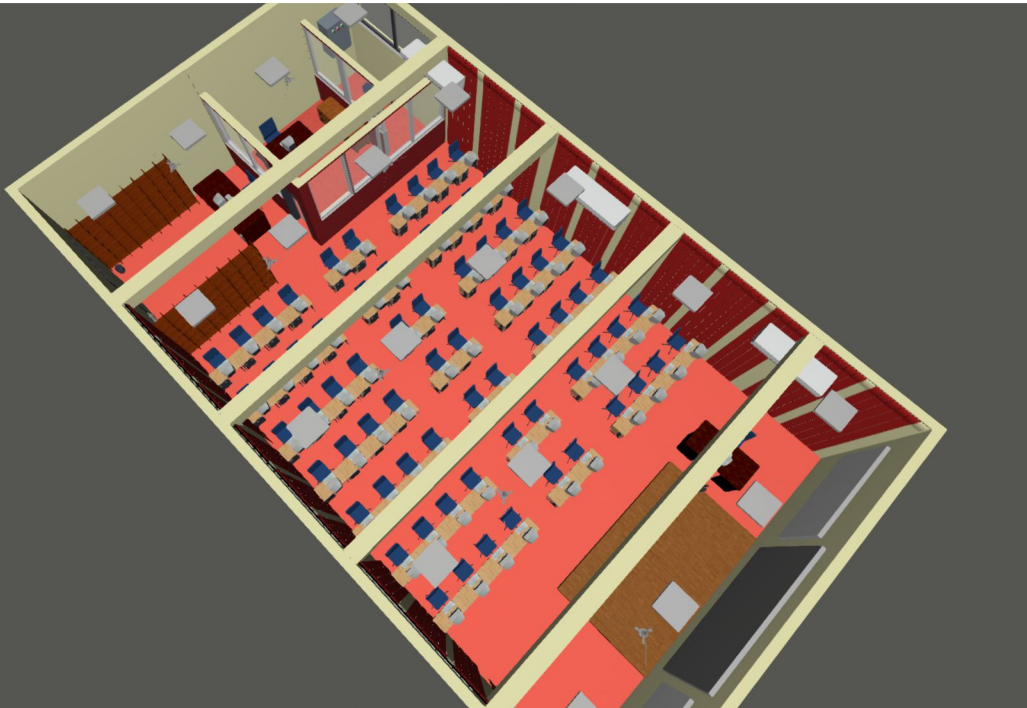
Contacts



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Description

This project illustrates the current lighting arrangement of the Computer Lab of Department of Electrical Engineering, University of Moratuwa. The project was implemented as an assignment given under module EE4410 - Electrical Services for Buildings.

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Luminaire list

Φ_{total} 67920 lm	P_{total} 1240.0 W	Luminous efficacy 54.8 lm/W
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pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
20	ASTZ	-	LVO10-4x14-031 Rastr HF	62.0 W	3396 lm	54.8 lm/W

Product data sheet

ASTZ - LVO10-4x14-031 Rastr HF



Article No.	-
P	62.0 W
Φ_{Lamp}	4800 lm
$\Phi_{\text{Luminaire}}$	3396 lm
η	70.75 %
Luminous efficacy	54.8 lm/W
CCT	3850 K
CRI	79

ЛВО10-4x14-031 Rastr HF

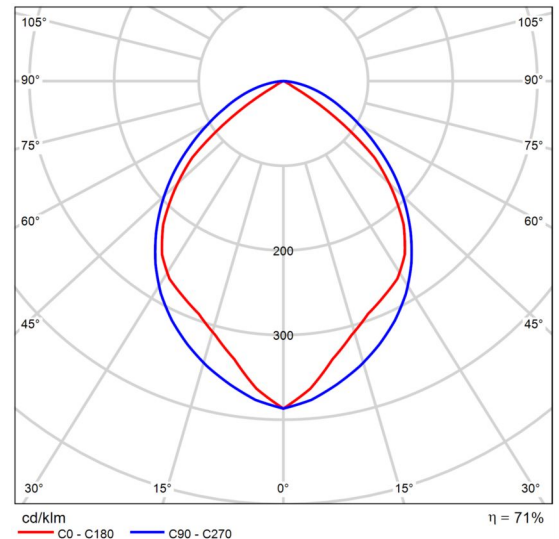
ТУ 3461-016-05014332-94

Предназначены для общего освещения общественных и иных помещений

220 В, 50 Гц, ЛЛ 14Вт Т5 G5, ЭПРА А2 (cosφ: 0,96), IP20, УХЛ4 Класс защиты от поражения электрическим током: I LxVxH=595x595x69 мм. Масса: 3,9 кг

Корпус из листовой стали, окрашен белой порошковой краской
Решетка из зеркального алюминия, устанавливается в корпус скрытыми пружинами

Установка: встраивается в подвесные потолки типа "Армстронг" (600x600 мм).



Polar LDC

Glare evaluation according to UGR												
p Ceiling		70	70	50	50	30	70	70	50	50	30	
p Walls		50	30	50	30	30	50	30	50	30	30	
p Floor		20	20	20	20	20	20	20	20	20	20	
Room size X Y		Viewing direction at right angles to lamp axis					Viewing direction parallel to lamp axis					
2H	2H	14.1	15.3	14.4	15.5	15.7	16.7	17.9	17.0	18.1	18.4	
	3H	13.9	15.0	14.2	15.2	15.5	17.8	18.9	18.2	19.2	19.4	
	4H	13.8	14.8	14.2	15.1	15.4	18.3	19.3	18.6	19.5	19.8	
	6H	13.8	14.7	14.1	15.0	15.3	18.5	19.5	18.9	19.8	20.1	
	8H	13.7	14.6	14.1	14.9	15.2	18.6	19.5	19.0	19.8	20.1	
4H	12H	13.7	14.5	14.0	14.9	15.2	18.6	19.5	19.0	19.8	20.1	
	2H	14.6	15.6	14.9	15.9	16.1	16.9	17.9	17.2	18.2	18.5	
	3H	14.5	15.3	14.8	15.6	16.0	18.2	19.0	18.6	19.4	19.7	
	4H	14.4	15.1	14.8	15.5	15.8	18.8	19.5	19.1	19.9	20.2	
	6H	14.3	15.0	14.7	15.3	15.7	19.2	19.8	19.6	20.2	20.6	
8H	12H	14.3	14.9	14.7	15.3	15.7	19.3	19.9	19.7	20.3	20.7	
	2H	14.2	14.8	14.7	15.2	15.6	19.3	19.9	19.8	20.3	20.7	
	4H	14.5	15.1	14.9	15.5	15.9	18.7	19.3	19.1	19.7	20.1	
	6H	14.4	14.9	14.9	15.4	15.8	19.2	19.7	19.6	20.1	20.5	
	8H	14.4	14.8	14.9	15.3	15.8	19.3	19.8	19.8	20.2	20.7	
12H	12H	14.4	14.7	14.9	15.2	15.7	19.4	19.8	19.9	20.3	20.8	
	4H	14.5	15.1	15.0	15.5	15.9	18.6	19.2	19.1	19.6	20.0	
	6H	14.5	14.9	14.9	15.3	15.8	19.1	19.6	19.6	20.0	20.5	
	8H	14.4	14.8	14.9	15.3	15.8	19.3	19.7	19.8	20.2	20.7	
Variation of the observer position for the luminaire distances S												
S = 1.0H		+0.6 / -1.1					+0.1 / -0.2					
S = 1.5H		+2.0 / -7.0					+0.4 / -0.7					
S = 2.0H		+3.3 / -10.7					+0.8 / -1.3					
Standard table		BK01					BK04					
Correction summand		-4.6					0.7					
Corrected glare indices referring to 4800lm Total luminous flux												

UGR diagram (SHR: 0.25)

Building 1

Luminaire list Φ_{total}

67920 lm

 P_{total}

1240.0 W

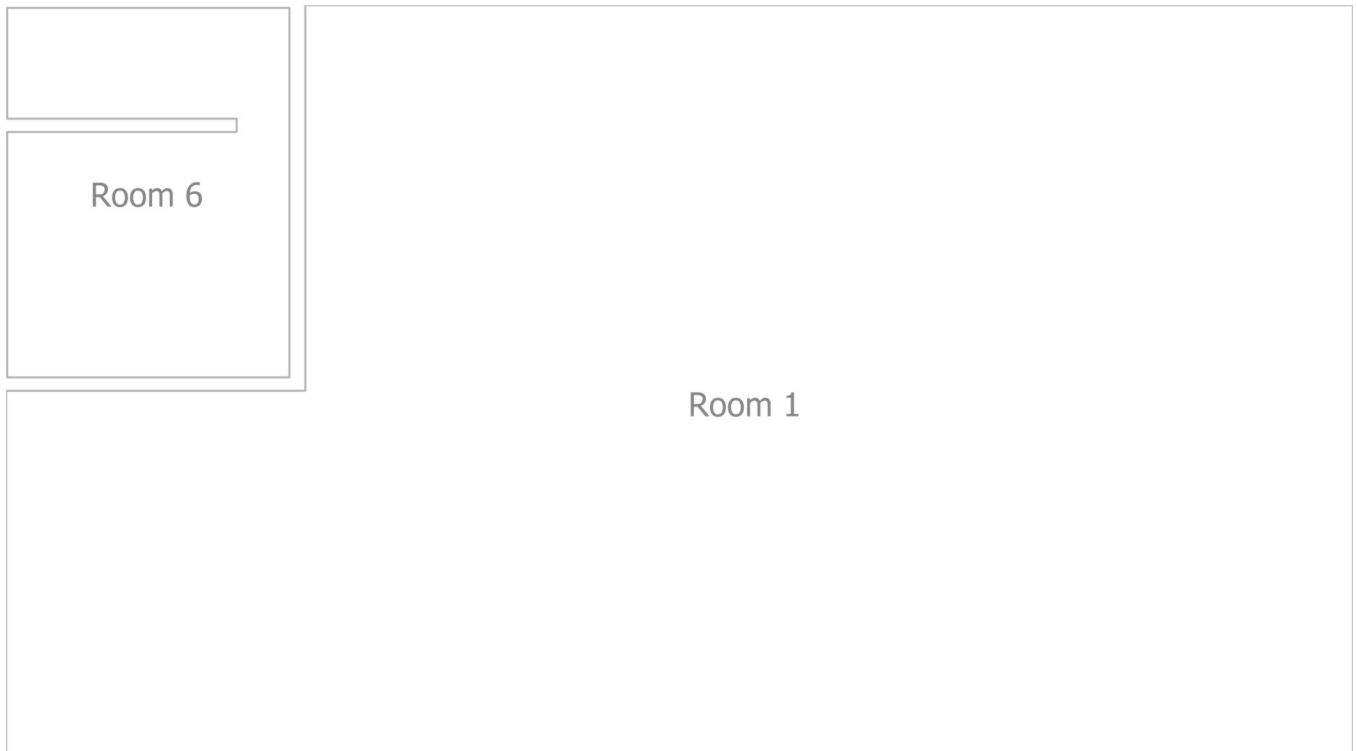
Luminous efficacy

54.8 lm/W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
20	ASTZ	-	LVO10-4x14-031 Rastr HF	62.0 W	3396 lm	54.8 lm/W

Building 1 · Storey 1 (Light scene 1)

Room list



Building 1 · Storey 1 (Light scene 1)

Room list

Room 1

P_{total} 1116.0 W	A_{Room} 141.83 m ²	Lighting power density 7.87 W/m ² = 2.68 W/m ² /100 lx (Space) 9.58 W/m ² = 3.26 W/m ² /100 lx (Working plane)	$\bar{E}_{\text{perpendicular (Working plane)}}$ 294 lx
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pcs.	Manufacturer	Article No.	Article name	P	$\Phi_{\text{Luminaire}}$
18	ASTZ	-	LVO10-4x14-031 Rastr HF	62.0 W	3396 lm

Room 6

P_{total} 124.0 W	A_{Room} 16.01 m ²	Lighting power density 7.75 W/m ² = 3.05 W/m ² /100 lx (Space) 10.73 W/m ² = 4.22 W/m ² /100 lx (Working plane)	$\bar{E}_{\text{perpendicular (Working plane)}}$ 254 lx
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pcs.	Manufacturer	Article No.	Article name	P	$\Phi_{\text{Luminaire}}$
2	ASTZ	-	LVO10-4x14-031 Rastr HF	62.0 W	3396 lm

Building 1 · Storey 1

Luminaire list Φ_{total}

67920 lm

 P_{total}

1240.0 W

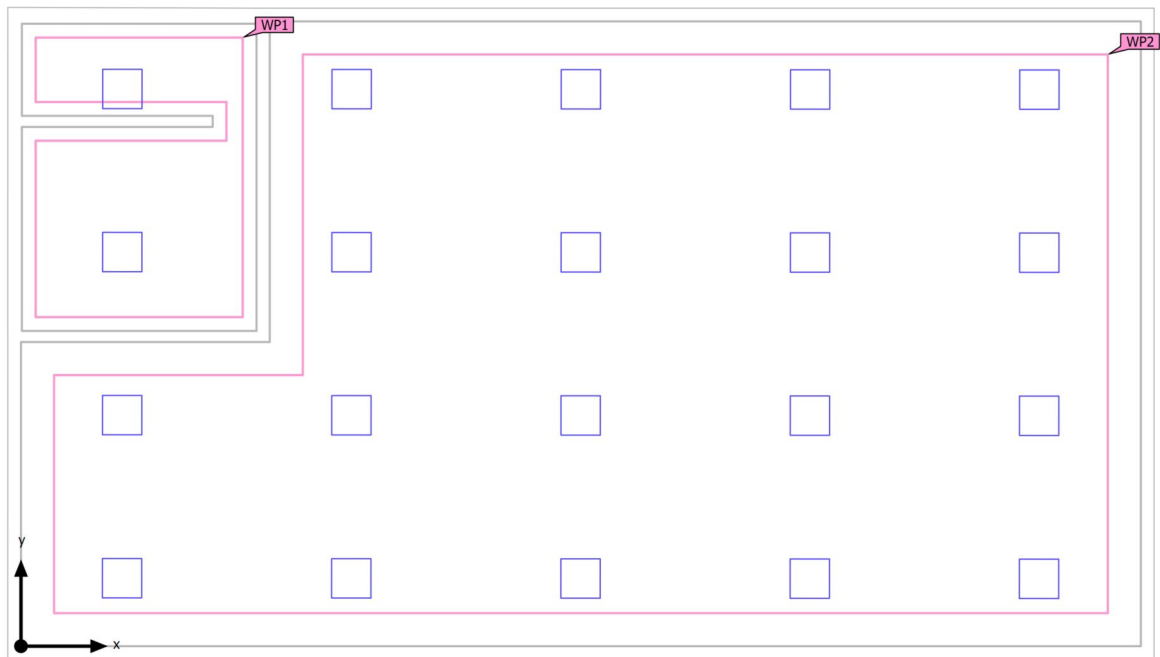
Luminous efficacy

54.8 lm/W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
20	ASTZ	-	LVO10-4x14-031 Rastr HF	62.0 W	3396 lm	54.8 lm/W

Building 1 · Storey 1 (Light scene 1)

Calculation objects



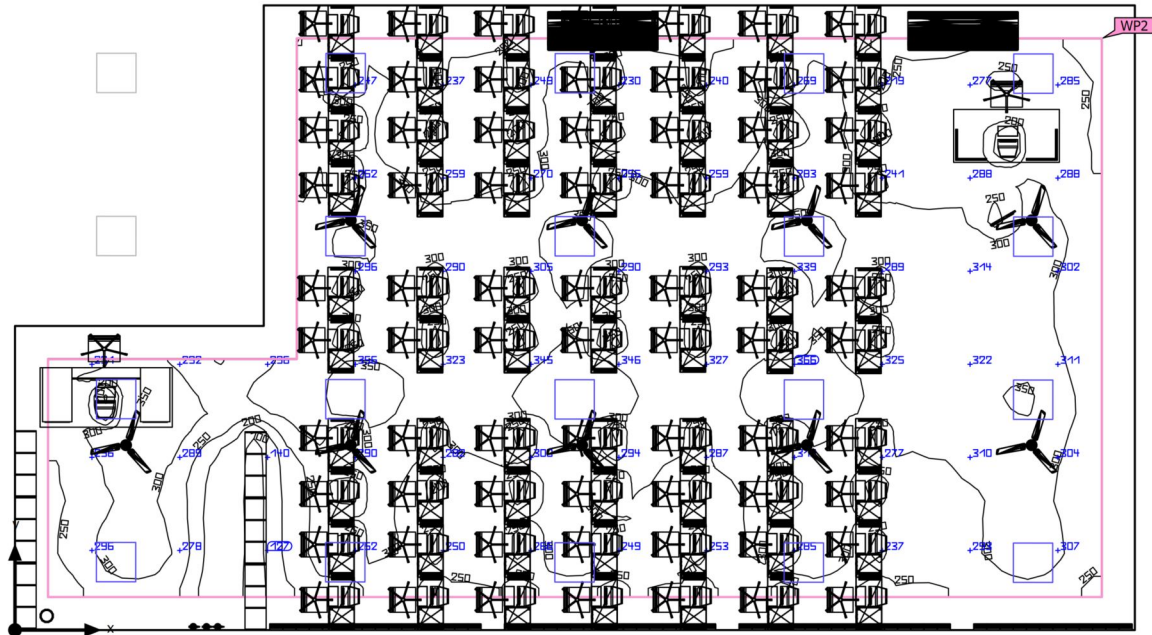
Building 1 · Storey 1 (Light scene 1)

Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane Perpendicular illuminance (adaptive) Height: 0.800 m, Wall zone: 0.209 m	254 lx (≥ 300 lx) ✗	187 lx	320 lx	0.74 (≥ 0.60) ✓	0.58	WP1
Working plane (Room 1) Perpendicular illuminance (adaptive) Height: 0.800 m, Wall zone: 0.500 m	294 lx (≥ 300 lx) ✗	52.3 lx	382 lx	0.18 (≥ 0.60) ✗	0.14	WP2

Building 1 · Storey 1 · Room 1 (Light scene 1)

Summary

Ground area	141.83 m ²		
Reflection factors	Ceiling: 60.5 %, Walls: 73.6 %, Floor: 28.8 %	Mounting height	3.727 m
Maintenance factor	0.80 (fixed)	Height _{Working plane}	0.800 m
		Wall zone _{Working plane}	0.500 m

Building 1 · Storey 1 · Room 1 (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	294 lx	≥ 300 lx	✗	WP2
	$U_o (g_1)$	0.18	≥ 0.60	✗	WP2
	Lighting power density	9.58 W/m ²	–		
		3.26 W/m ² /100 lx	–		
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	20	≤ 19	✗	
Energy estimation ⁽²⁾	Consumption	1484 kWh/a	max. 5000 kWh/a	✓	
Space	Lighting power density	7.87 W/m ²	–		
		2.68 W/m ² /100 lx	–		

(1) Based on a rectangular space of 16.942 m x 9.449 m and SHR of 0.25.

(2) Calculated using DIN:18599-4.

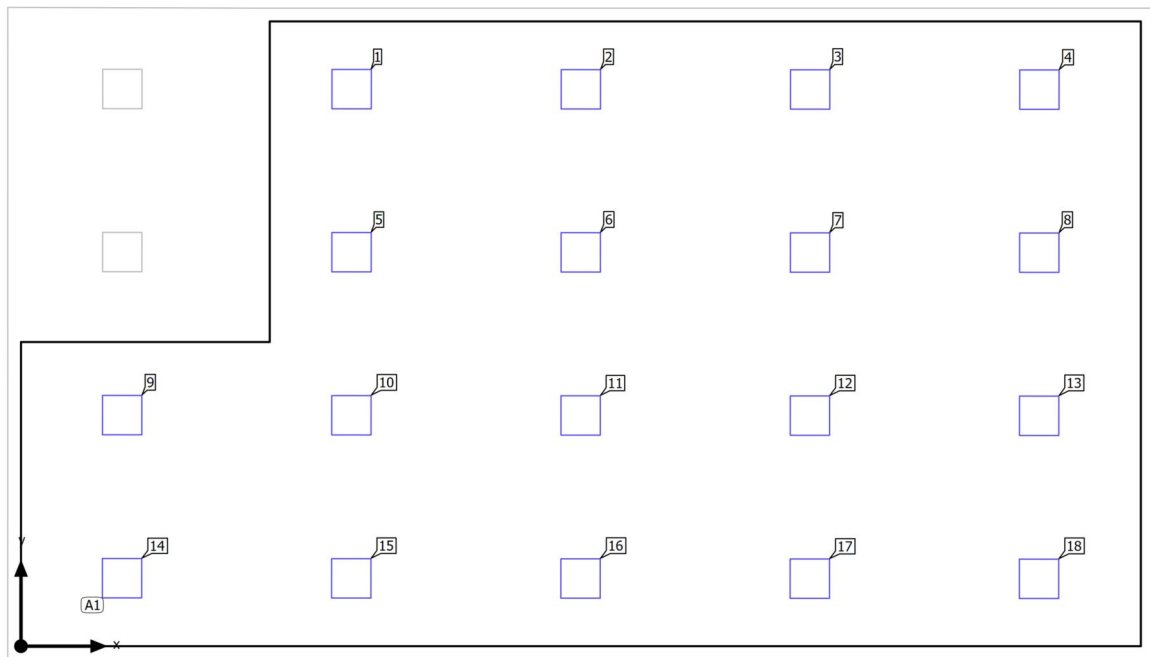
Utilisation profile: Educational premises - Educational buildings (44.11 Computer practice rooms (menu-driven))

Luminaire list

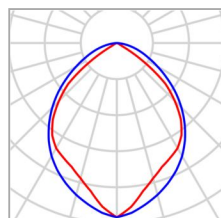
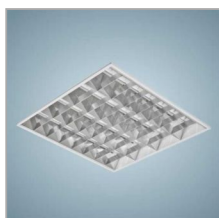
pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
18	ASTZ	-	LVO10-4x14-031 Rastr HF	20	62.0 W	3396 lm	54.8 lm/W

Building 1 · Storey 1 · Room 1

Luminaire layout plan



Building 1 · Storey 1 · Room 1

Luminaire layout plan

Manufacturer	ASTZ	P	62.0 W
Article No.	-	Φ _{Luminaire}	3396 lm
Article name	LVO10-4x14-031 Rastr HF		
Fitting	4x FL T5 G5 14W 1200lm		

20 x ASTZ LVO10-4x14-031 Rastr HF

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	1.528 m / 1.028 m / 3.727 m	5.002 m	8.424 m	3.727 m	1
X-direction	5 pcs., Centre - centre, 3.468 m	8.470 m	8.422 m	3.727 m	2
Y-direction	4 pcs., Centre - centre, 2.466 m	11.938 m	8.419 m	3.727 m	3
Arrangement	A1	15.407 m	8.417 m	3.727 m	4
		5.000 m	5.958 m	3.727 m	5
		8.468 m	5.956 m	3.727 m	6
		11.937 m	5.953 m	3.727 m	7
		15.405 m	5.951 m	3.727 m	8
		1.530 m	3.494 m	3.727 m	9
		4.998 m	3.492 m	3.727 m	10
		8.467 m	3.489 m	3.727 m	11
		11.935 m	3.487 m	3.727 m	12
		15.403 m	3.485 m	3.727 m	13

Building 1 · Storey 1 · Room 1

Luminaire layout plan

X	Y	Mounting height	Luminaire
1.528 m	1.028 m	3.727 m	14
4.997 m	1.026 m	3.727 m	15
8.465 m	1.023 m	3.727 m	16
11.933 m	1.021 m	3.727 m	17
15.402 m	1.018 m	3.727 m	18

Building 1 · Storey 1 · Room 1

Luminaire list Φ_{total}

61128 lm

 P_{total}

1116.0 W

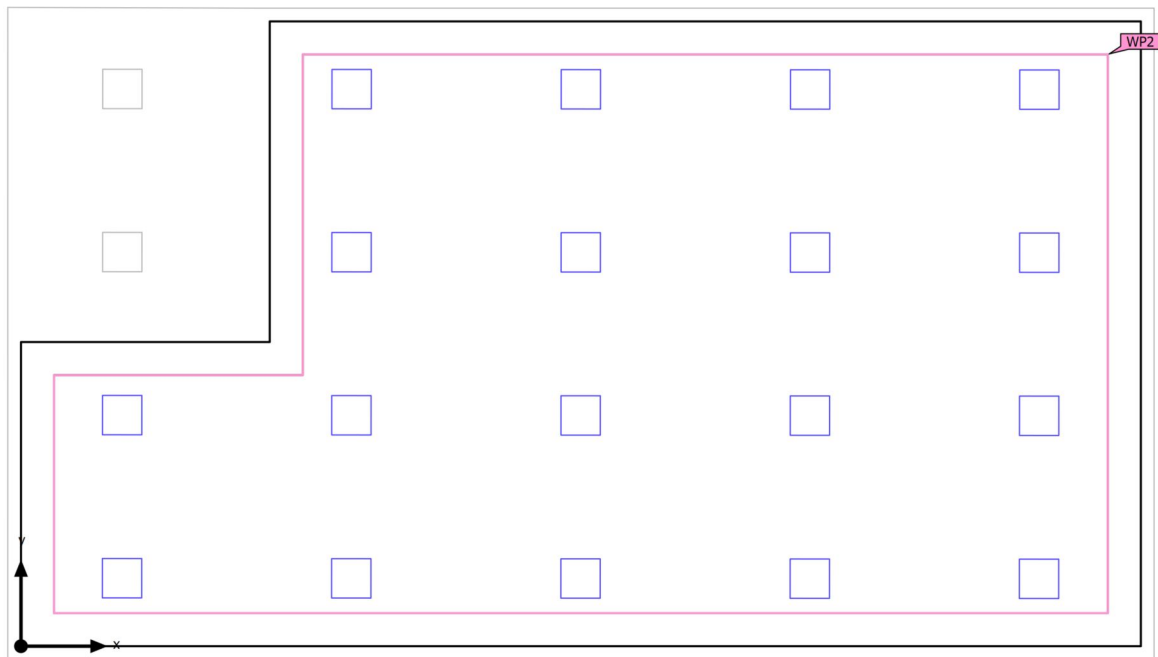
Luminous efficacy

54.8 lm/W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
18	ASTZ	-	LVO10-4x14-031 Rastr HF	62.0 W	3396 lm	54.8 lm/W

Building 1 · Storey 1 · Room 1 (Light scene 1)

Calculation objects



Building 1 · Storey 1 · Room 1 (Light scene 1)

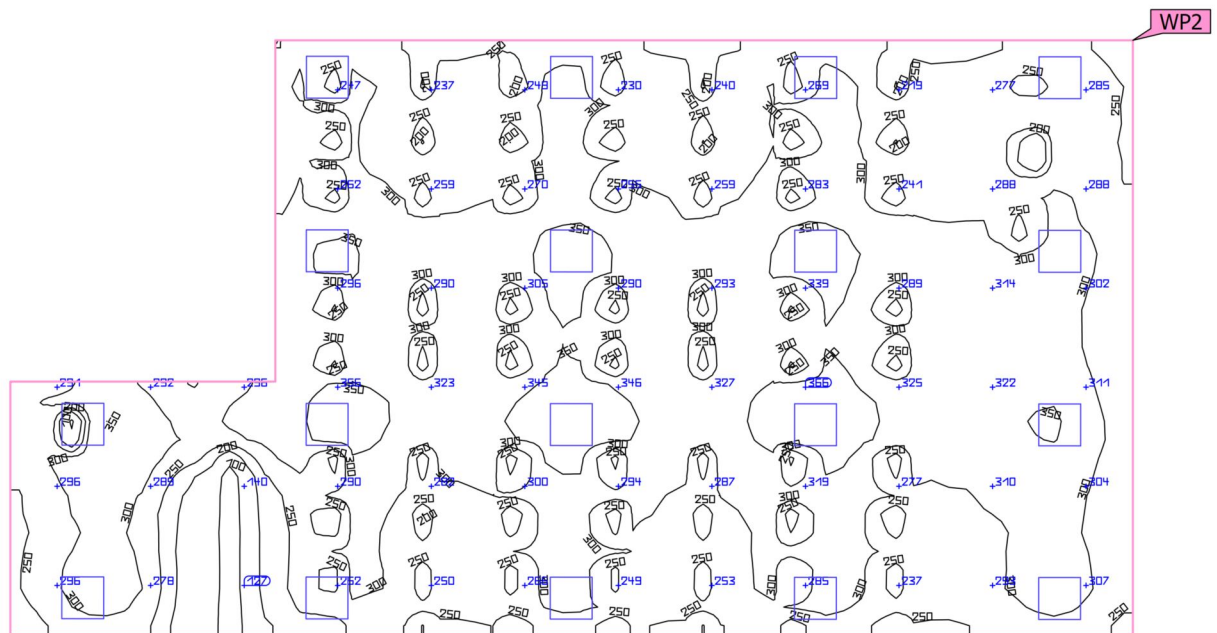
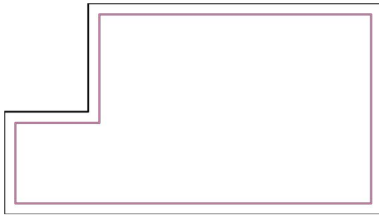
Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (Room 1)	294 lx	52.3 lx	382 lx	0.18	0.14	WP2
Perpendicular illuminance (adaptive)	(≥ 300 lx)			(≥ 0.60)		
Height: 0.800 m, Wall zone: 0.500 m	✗			✗		

Utilisation profile: Educational premises - Educational buildings (44.11 Computer practice rooms (menu-driven))

Building 1 · Storey 1 · Room 1 (Light scene 1)

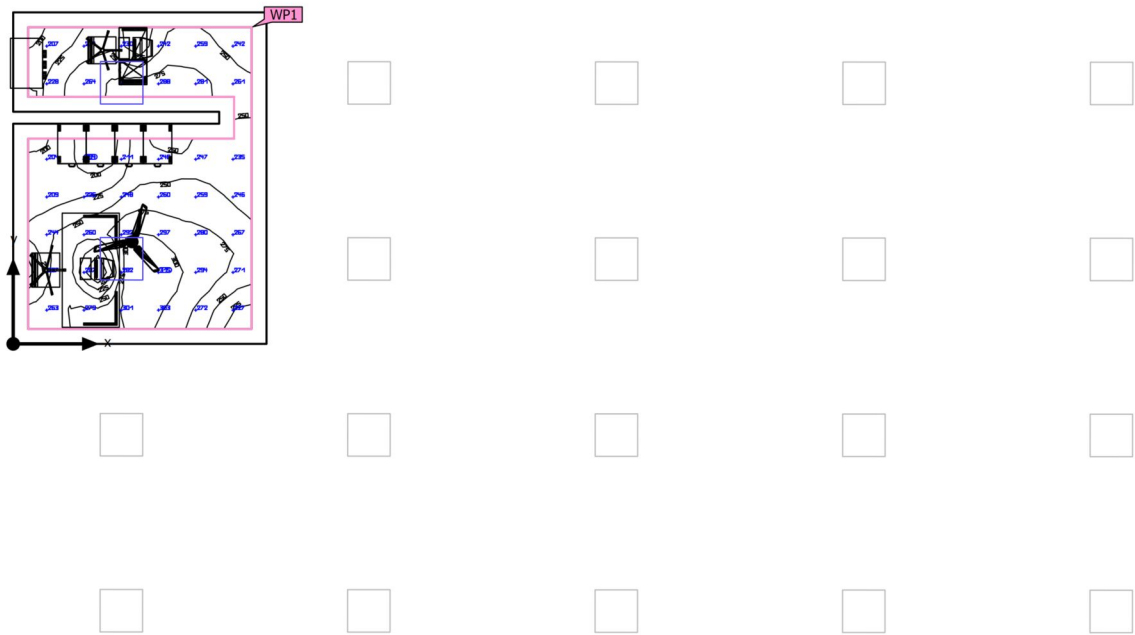
Working plane (Room 1)

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane (Room 1)	294 lx	52.3 lx	382 lx	0.18	0.14	WP2
Perpendicular illuminance (adaptive)	(≥ 300 lx)			(≥ 0.60)		
Height: 0.800 m, Wall zone: 0.500 m	✗			✗		

Utilisation profile: Educational premises - Educational buildings (44.11 Computer practice rooms (menu-driven))

Building 1 · Storey 1 · Room 6 (Light scene 1)

Summary



Ground area	16.01 m ²
Reflection factors	Ceiling: 72.5 %, Walls: 58.5 %, Floor: 23.0 %
Maintenance factor	0.80 (fixed)

Clearance height	3.658 m
Mounting height	3.727 m
Height _{Working plane}	0.800 m
Wall zone _{Working plane}	0.209 m

Building 1 · Storey 1 · Room 6 (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{\text{perpendicular}}$	254 lx	≥ 300 lx	✗	WP1
	$U_o (g_1)$	0.74	≥ 0.60	✓	WP1
	Lighting power density	10.73 W/m ²	–		
		4.22 W/m ² /100 lx	–		
Glare valuation ⁽¹⁾	$R_{UG, \text{max}}$	17	≤ 19	✓	
Energy estimation ⁽²⁾	Consumption	[99 - 165] kWh/a	max. 600 kWh/a	✓	
Space	Lighting power density	7.75 W/m ²	–		
		3.05 W/m ² /100 lx	–		

(1) Based on a rectangular space of 3.550 m x 4.645 m and SHR of 0.25.

(2) Calculated using DIN:18599-4.

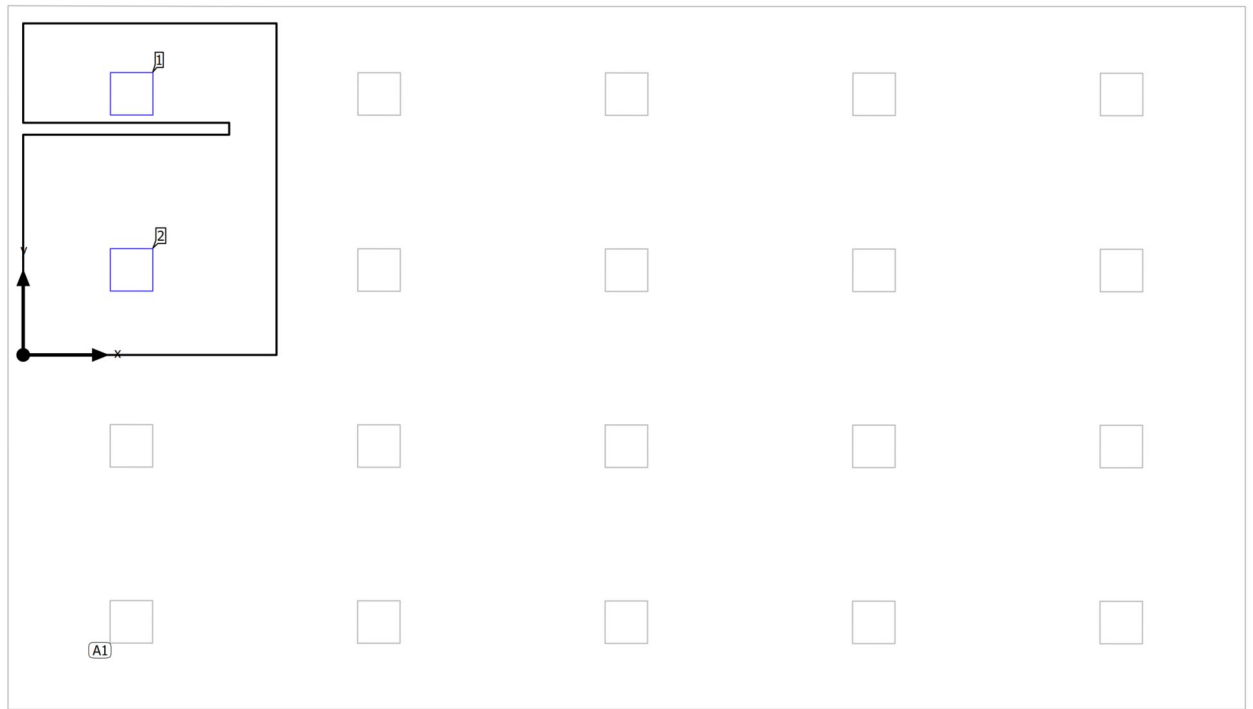
Utilisation profile: Educational premises - Educational buildings (44.11 Computer practice rooms (menu-driven))

Luminaire list

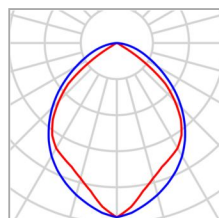
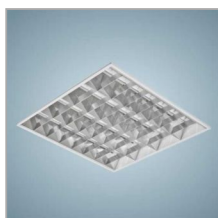
pcs.	Manufacturer	Article No.	Article name	R_{UG}	P	Φ	Luminous efficacy
2	ASTZ	-	LVO10-4x14-031 Rastr HF	17	62.0 W	3396 lm	54.8 lm/W

Building 1 · Storey 1 · Room 6

Luminaire layout plan



Building 1 · Storey 1 · Room 6

Luminaire layout plan

Manufacturer	ASTZ	P	62.0 W
Article No.	-	$\Phi_{\text{Luminaire}}$	3396 lm
Article name	LVO10-4x14-031 Rastr HF		
Fitting	4x FL T5 G5 14W 1200lm		

20 x ASTZ LVO10-4x14-031 Rastr HF

Type	Field Arrangement	X	Y	Mounting height	Luminaire
1st luminaire (X/Y/Z)	1.519 m / 1.193 m / 3.727 m	1.520 m	3.659 m	3.727 m	1
X-direction	5 pcs., Centre - centre, 3.468 m	1.519 m	1.193 m	3.727 m	2
Y-direction	4 pcs., Centre - centre, 2.466 m				
Arrangement	A1				

Building 1 · Storey 1 · Room 6

Luminaire list Φ_{total}

6792 lm

 P_{total}

124.0 W

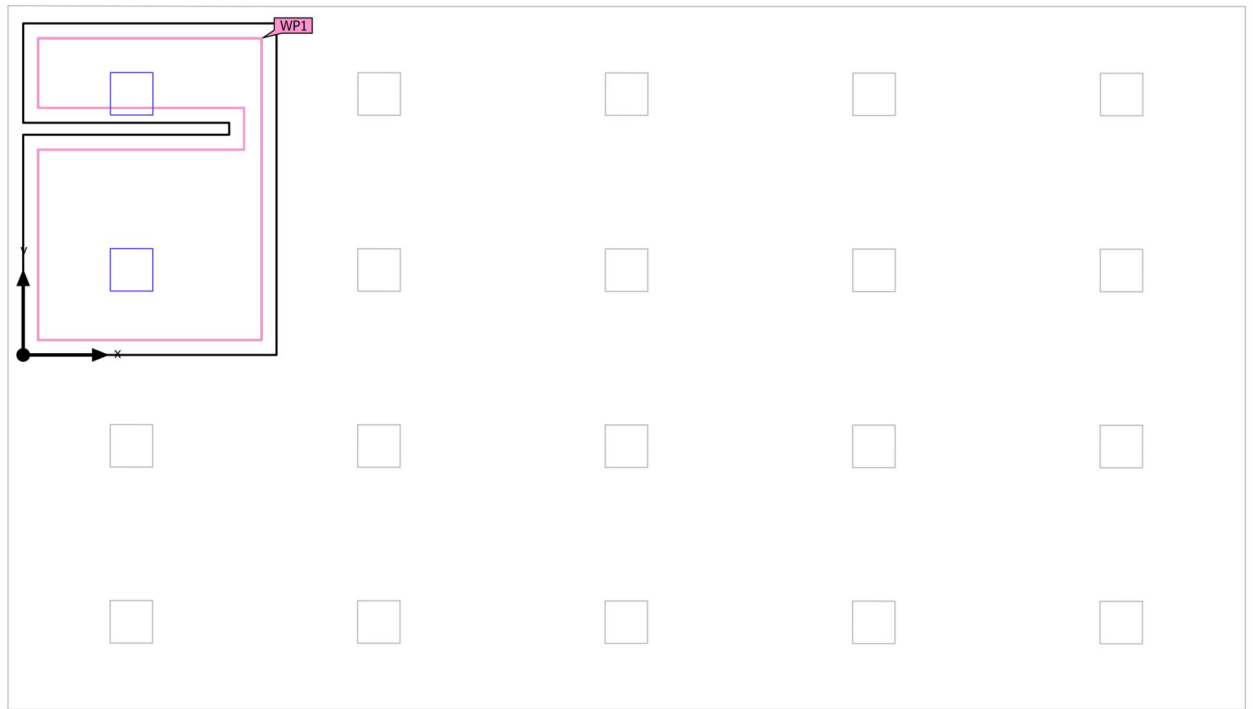
Luminous efficacy

54.8 lm/W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
2	ASTZ	-	LVO10-4x14-031 Rastr HF	62.0 W	3396 lm	54.8 lm/W

Building 1 · Storey 1 · Room 6 (Light scene 1)

Calculation objects



Building 1 · Storey 1 · Room 6 (Light scene 1)

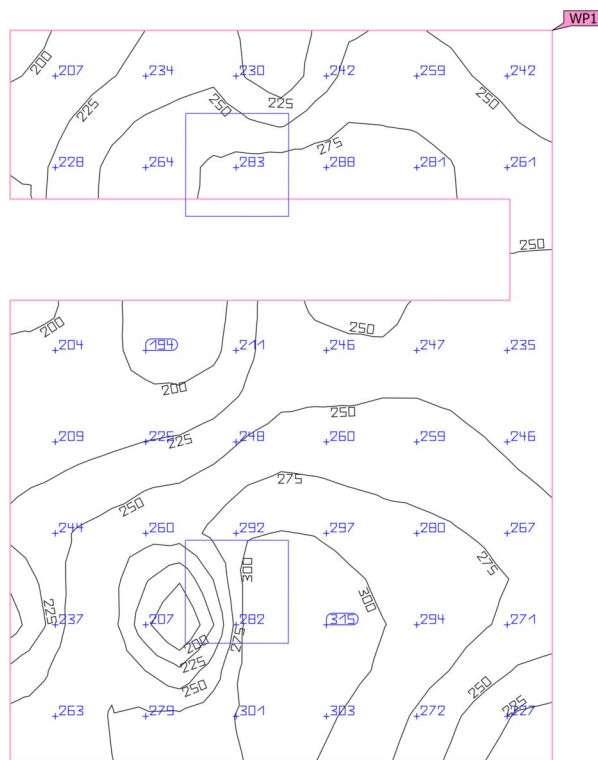
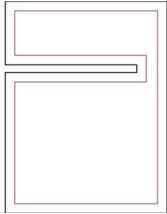
Calculation objects

Working planes

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane Perpendicular illuminance (adaptive) Height: 0.800 m, Wall zone: 0.209 m	254 lx (≥ 300 lx) ✗	187 lx	320 lx	0.74 (≥ 0.60) ✓	0.58	WP1

Utilisation profile: Educational premises - Educational buildings (44.11 Computer practice rooms (menu-driven))

Building 1 · Storey 1 · Room 6 (Light scene 1)

Working plane

Properties	$\bar{E}$ (Target)	E_{min}	E_{max}	$U_o (g_1)$ (Target)	g_2	Index
Working plane	254 lx	187 lx	320 lx	0.74	0.58	WP1
Perpendicular illuminance (adaptive)	(≥ 300 lx)			(≥ 0.60)		
Height: 0.800 m, Wall zone: 0.209 m	✗			✓		

Utilisation profile: Educational premises - Educational buildings (44.11 Computer practice rooms (menu-driven))